$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & 0 & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & -k^2\,\overset{(4)}{\kappa}_{3} & \frac{k^2\,\overset{(4)}{\kappa}_{4}}{2\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{k^2\,\overset{(4)}{\kappa}_{4}}{2\sqrt{2}} & \frac{Y_1\,k^2}{2} & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & k^2\,\overset{(4)}{\kappa}_{1} & \sqrt{2}\,k^2\,\overset{(4)}{\kappa}_{1} \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \sqrt{2}\,k^2\,\overset{(4)}{\kappa}_{1} & 2\,k^2\,\overset{(4)}{\kappa}_{1}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & 0 & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & 0 & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

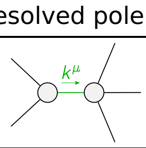
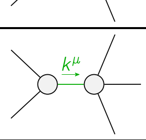
$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{Y_1\,k^2}{2\,\mathrm{Det}(1^+)} & -\frac{k^2\,\overset{(4)}{\kappa}_{4}}{2\sqrt{2}\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & -\frac{k^2\,\overset{(4)}{\kappa}_{4}}{2\sqrt{2}\,\mathrm{Det}(1^+)} & -\frac{k^2\,\overset{(4)}{\kappa}_{3}}{\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{1}{9\,k^2\,\overset{(4)}{\kappa}_{1}} & \frac{\sqrt{2}}{9\,k^2\,\overset{(4)}{\kappa}_{1}} \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{\sqrt{2}}{9\,k^2\,\overset{(4)}{\kappa}_{1}} & \frac{2}{9\,k^2\,\overset{(4)}{\kappa}_{1}}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & 0 & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

Abbreviations used in matrices
$Y_1 == 9\,\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4}\ \&\&\ \mathrm{Det}(1^+) == -\frac{1}{8}\,k^4\,(36\,\overset{(4)}{\kappa}_{1}\,\overset{(4)}{\kappa}_{3} + (2\,\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4})^2)$
Lagrangian
$\overset{(4)}{\kappa}_{1}\,\partial_{\beta}\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^{\chi}\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} - \overset{(4)}{\kappa}_{1}\,\partial_{\chi}\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^{\chi}\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_{3}\,\partial_{\beta}\mathcal{K}^{\alpha\beta\chi}\,\partial_{\delta}\mathcal{K}_{\alpha\chi}{}^{\delta} + \overset{(4)}{\kappa}_{4}\,\partial_{\alpha}\mathcal{K}^{\alpha\beta\chi}\,\partial_{\delta}\mathcal{K}_{\beta\chi}{}^{\delta} - \overset{(4)}{\kappa}_{3}\,\partial_{\beta}\mathcal{K}^{\alpha\beta\chi}\,\partial_{\delta}\mathcal{K}_{\chi\alpha}{}^{\delta} +$ $6\,\overset{(4)}{\kappa}_{1}\,\partial^{\chi}\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_{\delta}\mathcal{K}_{\chi\beta}{}^{\delta} + \frac{9}{2}\,\overset{(4)}{\kappa}_{1}\,\partial_{\alpha}\mathcal{K}^{\alpha\beta\chi}\,\partial_{\delta}\mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{2}\,\overset{(4)}{\kappa}_{3}\,\partial_{\alpha}\mathcal{K}^{\alpha\beta\chi}\,\partial_{\delta}\mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{2}\,\overset{(4)}{\kappa}_{4}\,\partial_{\alpha}\mathcal{K}^{\alpha\beta\chi}\,\partial_{\delta}\mathcal{K}^{\delta}{}_{\beta\chi}$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\ \mathcal{J}_{\alpha\beta\chi}$

Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1}_{0^-}{}^{\alpha\beta\chi} == 0$	1	$\partial_{\delta}\partial^{\alpha}\mathcal{J}^{\beta\chi\delta} + \partial_{\delta}\partial^{\alpha}\mathcal{J}^{\delta\beta\chi} + \partial_{\delta}\partial^{\beta}\mathcal{J}^{\chi\alpha\delta} + \partial_{\delta}\partial^{\chi}\mathcal{J}^{\alpha\beta\delta} + \partial_{\delta}\partial^{\chi}\mathcal{J}^{\delta\alpha\beta} + \partial_{\delta}\partial^{\delta}\mathcal{J}^{\beta\alpha\chi} ==$ $\partial_{\delta}\partial^{\alpha}\mathcal{J}^{\chi\beta\delta} + \partial_{\delta}\partial^{\beta}\mathcal{J}^{\alpha\chi\delta} + \partial_{\delta}\partial^{\beta}\mathcal{J}^{\delta\alpha\chi} + \partial_{\delta}\partial^{\chi}\mathcal{J}^{\beta\alpha\delta} + \partial_{\delta}\partial^{\delta}\mathcal{J}^{\alpha\beta\chi} + \partial_{\delta}\partial^{\delta}\mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}^{\#1}_{0^+} == 0$	1	$\partial_{\beta}\mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$
$\mathcal{J}^{\#1\alpha}_{1^-} == \mathcal{J}^{\#2\alpha}_{1^-}$	3	$\partial_{\chi}\partial^{\alpha}\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_{\chi}\partial^{\chi}\mathcal{J}^{\beta\alpha}{}_{\beta} == 0$
$\mathcal{J}^{\#1}_{2^-}{}^{\alpha\beta\chi} == 0$	5	$3\,\partial_{\epsilon}\partial_{\delta}\partial^{\chi}\partial^{\alpha}\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_{\epsilon}\partial^{\epsilon}\partial^{\chi}\partial^{\alpha}\mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\partial^{\beta}\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\partial^{\beta}\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\partial^{\beta}\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\partial^{\chi}\mathcal{J}^{\beta\alpha\delta} +$ $4\,\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\partial^{\chi}\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\partial^{\delta}\mathcal{J}^{\alpha\beta\chi} + 3\ \eta^{\beta\chi}\ \partial_{\phi}\partial^{\phi}\partial_{\epsilon}\partial^{\alpha}\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\ \eta^{\alpha\chi}\ \partial_{\phi}\partial^{\phi}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\delta\beta\epsilon} + 3\ \eta^{\beta\chi}\ \partial_{\phi}\partial^{\phi}\partial_{\epsilon}\partial^{\epsilon}\mathcal{J}^{\delta\alpha}{}_{\delta} ==$ $3\,\partial_{\epsilon}\partial_{\delta}\partial^{\chi}\partial^{\beta}\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_{\epsilon}\partial^{\epsilon}\partial^{\chi}\partial^{\beta}\mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\partial^{\alpha}\mathcal{J}^{\beta\chi\delta} + 4\,\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\partial^{\alpha}\mathcal{J}^{\chi\beta\delta} + 2\,\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\partial^{\alpha}\mathcal{J}^{\delta\beta\chi} + 2\,\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\partial^{\chi}\mathcal{J}^{\alpha\beta\delta} +$ $2\,\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\partial^{\delta}\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\partial^{\delta}\mathcal{J}^{\chi\alpha\beta} + 3\ \eta^{\alpha\chi}\ \partial_{\phi}\partial^{\phi}\partial_{\epsilon}\partial^{\beta}\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\ \eta^{\beta\chi}\ \partial_{\phi}\partial^{\phi}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\delta\alpha\epsilon} + 3\ \eta^{\alpha\chi}\ \partial_{\phi}\partial^{\phi}\partial_{\epsilon}\partial^{\epsilon}\mathcal{J}^{\delta\beta}{}_{\delta}$
$\mathcal{J}^{\#1\alpha\beta}_{2^+} == 0$	5	$3\,\partial_{\delta}\partial_{\chi}\partial^{\alpha}\mathcal{J}^{\chi\beta\delta} + 3\,\partial_{\delta}\partial_{\chi}\partial^{\beta}\mathcal{J}^{\chi\alpha\delta} + 2\ \eta^{\alpha\beta}\ \partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} == 2\,\partial_{\delta}\partial^{\beta}\partial^{\alpha}\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} + 3\,(\partial_{\delta}\partial^{\delta}\partial_{\chi}\mathcal{J}^{\alpha\beta\chi} + \partial_{\delta}\partial^{\delta}\partial_{\chi}\mathcal{J}^{\beta\alpha\chi})$
Total # constraint(s):	15	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$-\frac{48\,\overset{(4)}{\kappa}_{1}{}^2 + 8\,\overset{(4)}{\kappa}_{1}(2\,\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4}) + (2\,\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4})^2}{\overset{(4)}{\kappa}_{1}(36\,\overset{(4)}{\kappa}_{1}\,\overset{(4)}{\kappa}_{3} + (2\,\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4})^2)}$
	1	0	$-\frac{216\,\overset{(4)}{\kappa}_{1}{}^2 + 24\,\overset{(4)}{\kappa}_{1}(2\,\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4}) + (2\,\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4})^2}{\overset{(4)}{\kappa}_{1}(36\,\overset{(4)}{\kappa}_{1}\,\overset{(4)}{\kappa}_{3} + (2\,\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4})^2)}$

Resolved unitarity condition(s):	
	$(\overset{(4)}{\kappa}_{4} < 0\ \&\&\ ((\overset{(4)}{\kappa}_{3} < 0\ \&\&\ (\overset{(4)}{\kappa}_{1} < 0\ \|\|\ \overset{(4)}{\kappa}_{1} > \frac{-4\,\overset{(4)}{\kappa}_{3}{}^2 - 4\,\overset{(4)}{\kappa}_{3}\,\overset{(4)}{\kappa}_{4} - \overset{(4)}{\kappa}_{4}{}^2}{36\,\overset{(4)}{\kappa}_{3}}))\ \|\|\ (\overset{(4)}{\kappa}_{3} == 0\ \&\&\ \overset{(4)}{\kappa}_{1} < 0)\ \|\|$ $(0 < \overset{(4)}{\kappa}_{3} < -\frac{\overset{(4)}{\kappa}_{4}}{2}\ \&\&\ \frac{-4\,\overset{(4)}{\kappa}_{3}{}^2 - 4\,\overset{(4)}{\kappa}_{3}\,\overset{(4)}{\kappa}_{4} - \overset{(4)}{\kappa}_{4}{}^2}{36\,\overset{(4)}{\kappa}_{3}} < \overset{(4)}{\kappa}_{1} < 0)\ \|\|\ (\overset{(4)}{\kappa}_{3} > -\frac{\overset{(4)}{\kappa}_{4}}{2}\ \&\&\ \frac{-4\,\overset{(4)}{\kappa}_{3}{}^2 - 4\,\overset{(4)}{\kappa}_{3}\,\overset{(4)}{\kappa}_{4} - \overset{(4)}{\kappa}_{4}{}^2}{36\,\overset{(4)}{\kappa}_{3}} < \overset{(4)}{\kappa}_{1} < 0)))\ \|\|$ $(\overset{(4)}{\kappa}_{4} == 0\ \&\&\ ((\overset{(4)}{\kappa}_{3} < 0\ \&\&\ (\overset{(4)}{\kappa}_{1} < 0\ \|\|\ \overset{(4)}{\kappa}_{1} > -\frac{\overset{(4)}{\kappa}_{3}}{9}))\ \|\|\ (\overset{(4)}{\kappa}_{3} > 0\ \&\&\ -\frac{\overset{(4)}{\kappa}_{3}}{9} < \overset{(4)}{\kappa}_{1} < 0)))\ \|\|$ $(\overset{(4)}{\kappa}_{4} > 0\ \&\&\ ((\overset{(4)}{\kappa}_{3} < -\frac{\overset{(4)}{\kappa}_{4}}{2}\ \&\&\ (\overset{(4)}{\kappa}_{1} < 0\ \|\|\ \overset{(4)}{\kappa}_{1} > \frac{-4\,\overset{(4)}{\kappa}_{3}{}^2 - 4\,\overset{(4)}{\kappa}_{3}\,\overset{(4)}{\kappa}_{4} - \overset{(4)}{\kappa}_{4}{}^2}{36\,\overset{(4)}{\kappa}_{3}}))\ \|\|$ $(\overset{(4)}{\kappa}_{3} == -\frac{\overset{(4)}{\kappa}_{4}}{2}\ \&\&\ (\overset{(4)}{\kappa}_{1} < 0\ \|\|\ \overset{(4)}{\kappa}_{1} > 0))\ \|\|\ (-\frac{\overset{(4)}{\kappa}_{4}}{2} < \overset{(4)}{\kappa}_{3} < 0\ \&\&\ (\overset{(4)}{\kappa}_{1} < 0\ \|\|\ \overset{(4)}{\kappa}_{1} > \frac{-4\,\overset{(4)}{\kappa}_{3}{}^2 - 4\,\overset{(4)}{\kappa}_{3}\,\overset{(4)}{\kappa}_{4} - \overset{(4)}{\kappa}_{4}{}^2}{36\,\overset{(4)}{\kappa}_{3}}))\ \|\|$ $(\overset{(4)}{\kappa}_{3} == 0\ \&\&\ \overset{(4)}{\kappa}_{1} < 0)\ \|\|\ (\overset{(4)}{\kappa}_{3} > 0\ \&\&\ \frac{-4\,\overset{(4)}{\kappa}_{3}{}^2 - 4\,\overset{(4)}{\kappa}_{3}\,\overset{(4)}{\kappa}_{4} - \overset{(4)}{\kappa}_{4}{}^2}{36\,\overset{(4)}{\kappa}_{3}} < \overset{(4)}{\kappa}_{1} < 0)))$